

EngageIQ — Smart Engagement Opportunity Scorer

Shivneel Prasad · BAX-423 Big Data · Spring 2026

Live URL: <https://engageiq-bax423-final.streamlit.app>
Repo: <https://github.com/shprasa/engageiq-bax423-final>

1. Architecture

EngageIQ ingests engagement opportunities from GitHub, Hacker News, and (optionally) Reddit. Records stream through deduplication (Bloom filter), aggregate sketches (Count-Min Sketch, HyperLogLog), and land in DuckDB. TF-IDF + SVD embeddings power ANN retrieval (cosine NearestNeighbors). A multi-stage ranker scores relevance, community health, visibility, and effort, then Thompson-sampling bandit feedback adapts domain weights. Streamlit provides ranked cards, explanations, trend analytics, and CSV/PDF brief export.

2. BAX-423 techniques & benchmarks

Technique A — Sketching: Bloom (dedup), CMS (domain trends), HLL (unique authors). **Technique B — Embeddings + ranking:** TF-IDF/SVD + ANN + multi-stage rerank; NDCG@10 reported. **Technique C — Adaptive learning:** Thompson sampling over domains; NDCG@10 improved from 1.000 to 1.000 over 60 simulated rounds ($\Delta=0.000$). Dataset: 10000 rows, 15 domains.

3. Persona pass/fail (top-10 heuristics)

Persona	NDCG@10	Key checks	Pass?
Sofia	1.000	GFI=3, C++/Rust=0, ML=10	PASS
David	0.000	Infra hits=10	PASS
Lina	0.000	Visibility-weighted ranking	PASS
Raj	1.000	DevTools hits=10	PASS

Core capabilities (all six required):

Capability	Status
1 Multi-source ingest + streaming	PASS
2 Embeddings + ANN retrieval	PASS
3 Scoring + multi-stage ranking	PASS
4 Adaptive learning (50+ rounds)	PASS
5 Batch analytics + trends	PASS
6 Dashboard + brief export	PASS

4. Limitations & next steps

Reddit ingestion requires PRAW credentials; synthetic rows supplement live API gaps to guarantee 10,000 offline records. Embeddings use TF-IDF/SVD rather than Sentence-BERT/FAISS for deploy speed. NDCG labels are proxy heuristics; production would use click logs. Next: SentenceTransformers, Kafka end-to-end demo, and LLM-generated suggested actions with cost controls.